Building and Air Maintenance LLC
Commercial HVAC Restoration and Indoor Air Quality | Pine Grove, PA | (570) 900-5827 | buildingandair.com

BAM TECHNICAL GUIDE | FACILITY COMPLIANCE SERIES

ASHRAE 62.1 Section 8: The Operations and Maintenance Requirements Most Buildings Are Failing (and What Changed in 2025)

BAM Technical Team | August 2026 | buildingandair.com/blog
NADCA Certified | ASHRAE Member | OSHA-30 Certified | EPA Certified | PA COSTARS Vendor

> Everyone reads Section 6 for the ventilation rates. Almost nobody reads Section 8, which tells you what has to happen to filters, coils, drain pans, dampers, and outdoor air after the building is occupied. ASHRAE rewrote the whole table in 2025. Here is what it now requires.

Ask a facility director what ASHRAE 62.1 requires and you will hear about outdoor air rates: cfm per person, cfm per square foot, Table 6-1. That is Section 6, and it governs design. The section that governs the building you are running today is Section 8, Operations and Maintenance, and in most buildings we walk into, nobody has read it.

That matters more now than it did a year ago. ASHRAE published Standard 62.1-2025 in late 2025, and on September 30, 2025 approved Addendum h to 62.1-2022, which replaced the entire Section 8 maintenance table with a new one built around inspection, verification, and validation. If your preventive maintenance program was written to an older edition, or was never written to any edition, this is what the current standard expects.

Does Section 8 Apply to Your Building?

Section 8.1.1 of the 2022 edition says the section applies to buildings, ventilation systems, and components constructed or renovated after the adoption date of the section by your jurisdiction. Addendum h rewrites that to systems "placed into service," which drops the constructed-or-renovated limiter; once a jurisdiction adopts that text, Section 8 reads as applying to existing buildings in service.

Adoption runs through the mechanical code. The 2024 International Mechanical Code references ASHRAE 62.1-2022. The 2021 IMC, which Pennsylvania's Uniform Construction Code adopted effective January 1, 2026, references 62.1-2019. New Jersey, Maryland, and New York adopt an IMC edition statewide; Delaware adopts by county. In practice, a building permitted in the last decade in our territory was designed to somewhere between 62.1-2007 (Pennsylvania stayed on the 2009 codes until late 2018) and 62.1-2019, and the Section 8 tables in those editions are close cousins of the 2022 version.

If your building predates adoption, Table 8-1 is not enforceable against you today. It is still the benchmark that a commissioning agent, an insurer's engineer, an IAQ consultant investigating a complaint, or an expert witness will measure your program against. For healthcare facilities, ASHRAE 170 governs the clinical spaces, but 62.1 governs everything else on the campus and is referenced by 170 for outdoor air on mixed systems.

What Section 8 Requires Beyond the Table

Four requirements sit in the text and are missed as often as the table itself.

An O&M manual (Section 8.2). Written or electronic, kept on site or in a centrally accessible location for the working life of the equipment, updated as the system changes. It must contain the O&M procedures, ventilation system operating schedules and any changes to them, final design drawings, maintenance schedules based on manufacturer instructions, and the Table 8-1 inspection and maintenance requirements. Addendum h adds the maintenance tasks from ASHRAE/ACCA Standard 180.

Reevaluation on change of use (Section 8.1.2). When occupancy category, occupant density, or use changes in a way that no longer matches the design assumptions, the ventilation design, operation, and maintenance must be reevaluated and the manual updated. A conference room converted to an open office, or a storage room converted to a fitness space, triggers this.

Operation consistent with the manual (Section 8.3). Systems must be operated so that spaces receive Section 6 ventilation during periods of expected occupancy. An outdoor air damper closed for energy savings during occupied hours is a Section 8.3 violation.

The turnover package (Section 7.2.6). At construction close-out, the owner must receive and retain the O&M manual, control drawings and sequences, the air balance report, record drawings, and the design criteria. If you cannot find your balance report, this is the section that says you should have one.

Table 8-1 as Printed in 62.1-2022

The 2022 table is titled Minimum Maintenance Activity and Frequency for Ventilation System Equipment and Associated Components. Frequencies may be increased or decreased if the O&M manual says so. Condensed by frequency:

Frequency	Required activity
As necessary	Investigate the system for water intrusion or accumulation and correct it
Monthly	Treat open and closed cooling towers and evaporative condensers to limit microbial growth including Legionella; verify maintenance access to them is unobstructed
Quarterly	Check filter pressure drop and scheduled replacement date; check UV lamps; visually inspect humidifiers and dehumidifiers for fouling and microbial growth and verify RH control; verify maintenance access to equipment is unobstructed

Frequency	Required activity
Semiannually	Check cooling and heating coils for damage or leaks, clean, restore, or replace; visually inspect outdoor air intake louvers, bird screens, and mist eliminators and remove debris and biological material; maintain floor drains and trap primers in plenums; check belts, sheaves, VFDs, and IAQ-related controls
Annually	Verify operation of the outdoor air ventilation system and any dynamic minimum outdoor air controls; check filter fit and housing seal integrity; check drain pans, drain lines, and coils for biological growth and adjacent areas for unintended wetting; check heat exchange surfaces for fouling and restore; inspect for moisture carryover past the drain pan; check damper operation; inspect areas of moisture accumulation for biological growth and clean or disinfect; check condensate pumps, fan blades and housings, equipment panels, bearings, and exposed duct and pipe insulation
Every 5 years	Verify accuracy of permanently mounted outdoor air and DCV sensors, including CO2 sensors; verify the total outdoor air delivered by each air handler in minimum outdoor air mode against the design minimum within a 10 percent balancing tolerance, and correct it if short (units under 2,000 cfm exempt)

Two items stand out for anyone who manages coils. The standard expects coils to be checked and cleaned, restored, or replaced on a six-month cycle, and expects drain pans, drain lines, and coils to be checked for biological growth annually. In most institutional buildings we assess, the coils were cleaned at installation and never again. We wrote about what that does to static pressure and energy in how dirty coils increase static pressure and energy spend.

What Addendum h and the 2025 Edition Changed

Addendum h, approved by ASHRAE and ANSI on September 30, 2025, replaced Table 8-1 with a table titled Minimum Inspection, Verification, and Validation Activity and Frequency for Ventilation Systems and Associated Components. The committee's reasoning was that the old table duplicated maintenance tasks from ASHRAE/ACCA Standard 180, so the new table is built on inspection rather than maintenance, and Standard 180-2018 becomes a normative reference for the maintenance itself.

The new table distinguishes two kinds of tasks. "Verify" tasks (suffix a) can be done by visual inspection, documentation review, gauges, or BAS trend data without special tools. "Validate" tasks (suffix b) require calibrated instruments, adjustments, and specialized personnel. If a task cannot be verified, it must be validated regardless of the listed frequency.

Task	Frequency	What is new
Confirm whether the building is in a NAAQS nonattainment area; if so, plan to comply with outdoor air treatment requirements	Annually	New. EPA's 9 µg/m³ PM2.5 standard was upheld by the D.C. Circuit in June 2026; if EPA finalizes designations under it, more buildings will be pulled into MERV 11 outdoor air filtration
Inspect outdoor air intakes, exhaust openings, screens, louvers, and dampers for rain, snow, pests, growth, corrosion, and debris	Semiannually	Expanded scope, root cause required

Task	Frequency	What is new
Verify filter efficiency rating, thickness, fit, seal integrity, and replacement at maximum pressure drop	Quarterly, or at scheduled replacement if sooner	Fit and seal moved from annual to quarterly
Inspect drain pans, seals, traps, primers, pumps, and drain piping: pan under the coil, sloped, free of growth, draining freely, traps wet	Semiannually	Moved from annual to semiannual, and much more specific
Inspect humidifiers and dehumidifiers; validate humidity and dew point sensors	Semiannually; sensors every 3 years	Humidity sensor validation is new; humidifier inspection relaxed from quarterly to semiannual
Verify access is unobstructed for balancing, maintenance, sensors, and that access doors and panels function and seal	Semiannually	Relaxed from quarterly; access doors and seals now explicitly checked
Inspect plenums and ductwork for growth, corrosion, pests, and stored chemicals	Annually	New as a standalone task
Air balance verification: confirm airflows against design if use changed, verify displayed outdoor air flows meet design, verify exhaust exceeds supply where required, verify CO2 sensor calibration dates and DCV setup	Annually	Replaces the 5-year outdoor air check
Air balance validation: measure and adjust outdoor airflow on all units, spot check at least 20 percent of zones, rebalance exhaust for pressurization, calibrate CO2 and pressure sensors, clean and calibrate airflow sensors	Every 3 years	New. Outdoor air and CO2 sensor calibration moves from 5 years to 3; the 10 percent tolerance and the 2,000 cfm exemption are gone
Verify indoor humidity stays below the Section 5.12 limit	Annually	New
Verify directional airflow and building pressurization: zones positive or negative per design, flow toward exhaust, pressure sensors working	Annually	New
Inspect damper assemblies (seals, linkages, full travel, tight close, fail-safe), fan assemblies, zone controls, outdoor air at fan turndown, DCV response, and occupancy controls	Every 2 years	Consolidated under the variable-load and dynamic-reset task, more detailed, and stretched to two years (dampers were annual, belts and VFDs semiannual in 2022)
Verify the Legionella plan is implemented and maintained	Annually	New, linked to Section 5.20 and ASHRAE 188
Verify exhaust ventilation against Tables 6-2 and 6-3	Annually	New

The direction is unmistakable. The standard moved from "check the filter gauge quarterly" to "prove, annually, that the building is getting its design outdoor air and holding its design pressure relationships, and re-measure everything every three years." Addendum h is a published addendum to 62.1-2022 available free from ASHRAE; confirm whether your printing of the 2025 edition carries the revised table, and either way, the committee has told you where it is headed.

The Section 5 Requirements That Make Maintenance Possible

Several Section 5 design requirements exist specifically so that Section 8 can be carried out. When they are missing, maintenance gets skipped because it is physically impossible.

- **Access (Section 5.19).** Access doors or panels are required at the outdoor air intake plenum, mixed-air plenum, upstream face of every coil of four rows or fewer, both faces of every coil over four rows, air cleaners, drain pans and seals, fans, and humidifiers. A coil with no downstream access door will not be inspected on the downstream face, which is where the fouling is.
- **Drain pans (Section 5.7).** Sloped at least 0.125 inch per foot toward the outlet, outlet at the lowest point, P-trap or equivalent seal on draw-through units, pan extending downstream far enough to catch carryover.
- **Filtration upstream of wetted surfaces (Section 5.5).** MERV 8 minimum upstream of every cooling coil or other wetted device, a requirement that rose from MERV 6 in the 2013 edition. A coil running without a prefilter for a season will not come back with a quarterly filter change.
- **Humidity (Section 5.12).** Indoor dew point limited to 60°F whenever outdoor dew point is above 60°F, occupied and unoccupied.
- **Building exfiltration (Section 5.17).** In mechanically cooled buildings, total outdoor air intake must equal or exceed total exhaust under all load and dynamic reset conditions. A building running negative pulls humid air through the envelope and grows mold in the walls.
- **Intake separation (Table 5-1).** Cooling tower exhaust at least 25 feet from an outdoor air intake, cooling tower intake or basin 15 feet, loading docks and bus idling 25 feet, dumpsters 15 feet, plumbing vents 3 to 10 feet depending on height. Addendum h now requires an annual check that nothing has been built or moved inside those distances.
- **Legionella (Section 5.20).** The owner must receive written documentation of the HVAC water system design elements required by ASHRAE 188.

What We Find Against the Table

Section 8 expectation	Typical condition in a 15-year-old institutional building	Result
Coils checked semiannually, cleaned or restored	Cleaned at start-up, never since; downstream face inaccessible	Static pressure well above design, airflow down by double digits, latent capacity lost
Drain pans checked annually for growth	Standing water, biofilm, trap dry	Odor complaints, microbial amplification downstream, water in the insulation

Section 8 expectation	Typical condition in a 15-year-old institutional building	Result
Outdoor air verified annually, validated every 3 years	Last balance report at commissioning; economizer minimum overridden	CO2 above 1,000 ppm in afternoon classes and meetings; complaints logged as "stuffy"
Filter fit and seal checked quarterly	Filters undersized for the rack, gasket collapsed, bypass at the corners	Downstream coil fouling, MERV rating meaningless
Damper assemblies inspected every 2 years	Outdoor air damper linkage seized; actuator drives, blades do not move	Building negative, humidity high, or ventilation zero
Sensors validated every 3 years	CO2 sensors never calibrated since installation	DCV starving the space or wasting energy, no one knows which
O&M manual on site	Nobody can find it	No design basis to verify against

Every row is correctable without replacing equipment. Coil restoration returns heat transfer and airflow to near-original performance; drain pan and duct cleaning remove the growth; rebalancing and damper repair restore outdoor air. The bigger point is that each of these is a documented finding under the current standard, and the documentation is what protects you when a complaint, an insurance claim, or a survey arrives.

Frequently Asked Questions

What does ASHRAE 62.1 Section 8 require?

An operations and maintenance manual, operation of the ventilation system consistent with that manual during occupied hours, reevaluation whenever the building's use changes, and the inspection and maintenance activities and frequencies in Table 8-1.

How often does ASHRAE 62.1 require coils to be cleaned?

The 2022 edition's Table 8-1 requires coils to be checked semiannually and cleaned, restored, or replaced as needed, with an annual check of drain pans, drain lines, and coils for biological growth. Addendum h moves the maintenance task to ASHRAE/ACCA Standard 180 and requires a semiannual inspection of the drain pan assembly.

How often must outdoor air be verified?

The 2022 table required verification of total outdoor air per air handler every five years within a 10 percent tolerance. Addendum h (2025) requires annual verification and a full test-and-balance validation of every unit every three years, with no tolerance clause and no small-unit exemption.

Is ASHRAE 62.1-2025 adopted in Pennsylvania?

No. Pennsylvania's Uniform Construction Code adopted the 2021 International Codes effective January 1, 2026, and the 2021 IMC references ASHRAE 62.1-2019. The 2024 IMC references 62.1-2022. No published model code references the 2025 edition yet.

Does 62.1 Section 8 apply to hospitals?

ASHRAE 170 governs the clinical spaces, and it references 62.1 for outdoor air on systems serving both 170 and 62.1 spaces. Administrative, support, and campus buildings fall under 62.1 directly, and The Joint Commission's physical environment requirements expect utility system maintenance to be defined, performed, and documented regardless of which standard sets the frequencies.

What Happens Next

If your PM program cannot produce a coil inspection record, a current balance report, or a drain pan check from the last twelve months, it does not meet the standard your building was designed to.

1. **Request a Section 8 Compliance Assessment.** We inspect every air handler against Table 8-1 (both versions), measure coil static pressure, outdoor airflow, and pressure relationships, and photograph the conditions.
2. **Receive a gap report you can attach to your O&M manual.** Each item mapped to the section number, the required frequency, and the corrective action.
3. **Restore, then put it on a schedule.** Coil restoration, drain pan and duct cleaning, damper and sensor correction, rebalancing, and a written preventive maintenance program that follows the new inspection, verification, and validation frequencies.

Contact us today to request your Section 8 Compliance Assessment, or call (570) 900-5827.

RELATED SERVICES

- AHU & Coil Restoration
- Preventive Maintenance Programs
- NADCA-Aligned Duct Cleaning
- Commercial Indoor Air Quality
- Education (K-12 & Higher Ed)

RELATED ARTICLES

- ASHRAE 62.1 and 170 Explained for Facility Directors
- A Facility Director's Guide to IAQ Technical Standards: From ASHRAE 62.1 to PM2.5
- The Hidden Cost of Deferred HVAC Maintenance in Institutional Facilities

BAM Technical Team · 25+ Years in Institutional HVAC

Our articles are written by field technicians and engineers with decades of hands-on experience restoring HVAC systems in hospitals, universities, data centers, and manufacturing facilities. Building and Air Maintenance LLC is a NADCA-certified commercial HVAC restoration and indoor air quality contractor serving Pennsylvania, New Jersey, Maryland, Delaware, and southern New York. Request a site visit at buildingandair.com/contact or call (570) 900-5827.